LAB – 5

UML USE CASE MODELING

PROJECT REPORT

AI-BASED INTERVIEW PREPARATION PLATFORM

Actors, Use Cases, UML Use Case Diagram and Use Case Narratives

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1. Introduction

UML Use Case Modeling is used to represent the functional interaction between users or external systems and the proposed software. For the AI-Based Interview Preparation Platform, the use case model identifies the main actors and the services they expect from the system. The model is based on the requirements identified in the previous requirements engineering labs.

2. System Overview

The AI-Based Interview Preparation Platform is a web-based application that helps students and job seekers practice technical and HR interviews. A candidate can register, manage a profile, select a target job role and interview type, start an AI-assisted mock interview, answer questions, receive AI-based evaluation, view scores and feedback, and track preparation progress. An administrator manages users and interview content, while an AI/LLM service supports question generation and answer evaluation.

3. Actors Identification

Actor	Type	Description
Candidate	Primary / Human	Uses the platform to prepare for interviews, take mock interviews and view performance.
Administrator	Primary / Human	Manages users, questions and interview categories.
AI / LLM Service	External System	Generates interview questions and assists with response evaluation.

4. Use Case Identification

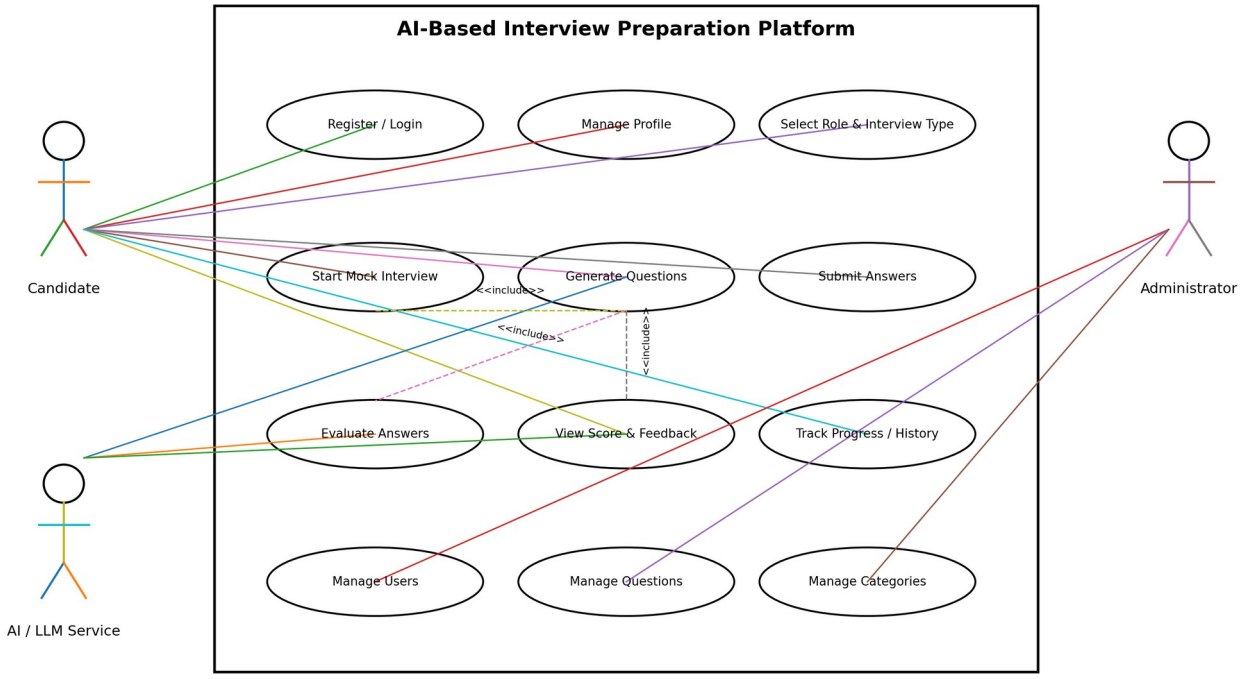
ID	Use Case	Primary Actor	Purpose
UC-01	Register / Login	Candidate	Create an account and securely access the platform.
UC-02	Manage Profile	Candidate	Maintain skills, education, experience and target role information.
UC-03	Select Role & Interview Type	Candidate	Choose job role and interview category for personalized preparation.
UC-04	Start Mock Interview	Candidate	Begin an interview practice session.
UC-05	Generate Questions	Candidate / AI Service	Generate or retrieve relevant questions based on role and difficulty.
UC-06	Submit Answers	Candidate	Provide responses to interview questions.
UC-07	Evaluate Answers	AI Service	Analyze responses and produce evaluation data.
UC-08	View Score & Feedback	Candidate	View performance score, strengths, weaknesses and suggestions.
UC-09	Track Progress / History	Candidate	Review previous attempts and performance trends.
UC-10	Manage Users	Administrator	View, update or manage platform users.
UC-11	Manage Questions	Administrator	Add, edit or remove interview questions.
UC-12	Manage Categories	Administrator	Manage interview categories,

			roles and question classifications.

5. UML Use Case Diagram

The following UML use case diagram shows the major actors and their interactions with the AI-Based Interview Preparation Platform.

UML Use Case Diagram



6. Use Case Relationships

Relationship	Use Cases	Explanation
Association	Actors ↔ Use Cases	Shows direct interaction between an actor and a system function.
<<include>>	Start Mock Interview → Generate Questions	Question generation is a required part of starting a personalized interview.
<<include>>	Start Mock Interview → Submit Answers	The interview session requires candidate responses.
<<include>>	Submit Answers → Evaluate Answers	Responses are evaluated to produce performance results.

7. Detailed Use Case Narratives

UC-01 – Register / Login

Field	Description
Primary Actor	Candidate
Description	The candidate enters registration or login credentials. The system validates the information and creates/authenticates the account. On successful authentication, the candidate is

	redirected to the dashboard.
Preconditions	Candidate has access to the platform. Required credentials are available.
Postconditions	Account created successfully. Candidate is authenticated and dashboard is displayed.
Alternative / Exception Flows	Invalid or incomplete credentials. Existing email during registration. Authentication service unavailable.

UC-02 – Manage Profile

Field	Description
Primary Actor	Candidate
Description	The candidate opens the profile section, enters or updates education, skills, experience, target role and other relevant information, and saves the changes.
Preconditions	Candidate is logged in.
Postconditions	Profile information is validated and stored. Updated profile is displayed.
Alternative / Exception Flows	Invalid profile data. Database update failure.

UC-03 – Select Role & Interview Type

Field	Description
Primary Actor	Candidate
Description	The candidate selects a target job role, interview type and difficulty level. The system stores the selection and uses it to personalize the interview.
Preconditions	Candidate is logged in. Available roles and types exist.
Postconditions	Interview configuration is saved successfully.
Alternative / Exception Flows	Unsupported role. Missing required selection.

UC-04 – Start Mock Interview

Field	Description
Primary Actor	Candidate
Description	The candidate chooses the interview configuration and starts a session. The system creates an interview session and prepares the required questions.
Preconditions	Candidate is logged in. Interview configuration is available.
Postconditions	Interview session is created. First question is displayed.
Alternative / Exception Flows	AI service unavailable. Session creation failure.

UC-05 – Generate Questions

Field	Description
Primary Actor	Candidate / AI Service
Description	The system sends relevant profile and interview parameters to the AI service or question repository. Questions are generated/retrieved and returned to the interview session.
Preconditions	Interview session has valid role/type/difficulty information.
Postconditions	Relevant questions are available to the candidate.
Alternative / Exception Flows	AI API failure. Rate limit. No suitable questions returned.

UC-06 – Submit Answers

Field	Description
Primary Actor	Candidate
Description	The candidate provides an answer for each question. The system validates and stores the response and moves to the next question until the session is completed.
Preconditions	An active interview session exists. A question is displayed.
Postconditions	Answer is stored against the correct question. Next question or

	completion screen is shown.
Alternative / Exception Flows	Empty/invalid answer. Network or server failure.

UC-07 – Evaluate Answers

Field	Description
Primary Actor	AI / LLM Service
Description	After answers are submitted, the system sends appropriate response data for analysis. The AI service returns evaluation information such as answer quality, strengths, weaknesses and suggested improvements.
Preconditions	Interview responses are available. AI service is available.
Postconditions	Evaluation is generated and stored. Score/feedback data is available.
Alternative / Exception Flows	AI service timeout. Invalid response from AI service.

UC-08 – View Score & Feedback

Field	Description
Primary Actor	Candidate
Description	The candidate opens the result page after completing an interview. The system displays the score, strengths, weaknesses and personalized recommendations.
Preconditions	Interview has been evaluated.
Postconditions	Candidate can view complete results.
Alternative / Exception Flows	Results not available. Temporary server/database failure.

UC-09 – Track Progress / History

Field	Description
Primary Actor	Candidate
Description	The candidate opens the progress dashboard. The system retrieves previous interview attempts and displays scores, history and performance trends.
Preconditions	Candidate is logged in. Previous attempts may exist.
Postconditions	History and progress information is displayed.
Alternative / Exception Flows	No previous attempts. Data retrieval failure.

UC-10 – Manage Users

Field	Description
Primary Actor	Administrator
Description	The administrator opens the user management module and performs permitted actions such as viewing, updating, activating or managing user records.
Preconditions	Administrator is authenticated and authorized.
Postconditions	Requested user-management action is completed and recorded.
Alternative / Exception Flows	Unauthorized access. Invalid user record.

UC-11 – Manage Questions

Field	Description
Primary Actor	Administrator
Description	The administrator adds, edits or removes interview questions and assigns appropriate categories or difficulty levels.
Preconditions	Administrator is authenticated. Question management permission is available.
Postconditions	Question content is saved or updated.
Alternative / Exception Flows	Invalid question data. Database failure.

UC-12 – Manage Categories

Field	Description
Primary Actor	Administrator
Description	The administrator manages job roles, interview types and question categories so that the platform maintains organized and relevant interview content.
Preconditions	Administrator is authenticated.
Postconditions	Category changes are saved and available to the platform.
Alternative / Exception Flows	Duplicate/invalid category. Database failure.

8. Main Candidate Scenario

Register/Login → Manage Profile → Select Role & Interview Type → Start Mock Interview → Generate Questions → Submit Answers → Evaluate Answers → View Score & Feedback → Track Progress / History

9. Main Administrator Scenario

Admin Login → Manage Users → Manage Questions → Manage Categories → Maintain Platform Content

10. Use Case Model Summary

The use case model provides a clear functional view of the proposed platform. Candidates are the main users and interact with authentication, profile, interview, evaluation, feedback and progress features. The administrator maintains the platform's users and content. The external AI/LLM service supports question generation and answer evaluation. These use cases can be used as a basis for detailed design, development and testing.

11. Conclusion

UML Use Case Modeling for the AI-Based Interview Preparation Platform successfully identifies the system actors and major functional interactions. The use case diagram and narratives describe how candidates, administrators and the AI service interact with the system. The model provides a structured representation of the approved requirements and can be used as a foundation for subsequent design, implementation and testing activities.